

## §2.3

# Multiplication and Division Problem Types

---

Four structures, and the unknown that names each subtype

# Today's Goal

---

After completing this section, you should be able to:

- Differentiate between the different types of multiplication and division problems based on context
- Understand how the different types of multiplication and division problems influence the models and strategies used

## **LT.5 Multiplication and Division Problem Types**

I can identify by name and create different types of story problems (e.g., Equal Groups Product Unknown) related to multiplication and division.

# Four Types, Three Subtypes Each

Table 2.3.1: Multiplication and Division Problem Types

| Type         | Subtypes       |                    |                    |
|--------------|----------------|--------------------|--------------------|
| Equal Groups | Multiplication | Partitive Division | Quotative Division |
| Array        | Multiplication | Partitive Division | Quotative Division |
| Area         | Multiplication | Partitive Division | Quotative Division |
| Compare      | Multiplication | Partitive Division | Quotative Division |

Every type has three subtypes, and the subtype is named by **which quantity is unknown**.

# Equal Groups Problems

Definition 2.3.2

An **Equal Groups problem** involves a set of objects divided into groups in which each group has the same number of objects.

Components: **number of groups**, **group size**, **product** (the total across all groups).

|                         |   |                         |   |                         |
|-------------------------|---|-------------------------|---|-------------------------|
| <b>number of groups</b> | × | <b>group size</b>       | = | <b>product</b>          |
| <b>product</b>          | ÷ | <b>number of groups</b> | = | <b>group size</b>       |
| <b>product</b>          | ÷ | <b>group size</b>       | = | <b>number of groups</b> |

# The Three Equal Groups Subtypes

Table 2.3.3: Equal Groups Problem Subtypes

| Equal Groups<br>(Multiplication)                                                     | Equal Groups (Partitive<br>Division)                                                  | Equal Groups<br>(Quotative Division)                                          |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| There are three bags with five apples in each bag. How many apples are there in all? | If 15 apples are shared equally into three bags, how many apples will be in each bag? | If 15 apples are to be packed five apples to a bag, how many bags are needed? |
| $3 \times 5 = ?$                                                                     | $3 \times ? = 15$                                                                     | $? \times 5 = 15$                                                             |
|                                                                                      | $15 \div 3 = ?$                                                                       | $15 \div 5 = ?$                                                               |

The student videos we will watch are from *Children's Mathematics* (Cognitively Guided Instruction, Thomas Carpenter and colleagues).



# Grandma's Cookies

Grandma has 3 plates with 6 cookies on every plate. How many cookies does Grandma have?

- a. **Face Partner A:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.  
**Face Partner B:** How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b. ▶ **Play: a student solves this problem** [smarturl.it/cm4.1](http://smarturl.it/cm4.1)
- c. ▶ **Play: another student solves a similar problem** [smarturl.it/cm4.5](http://smarturl.it/cm4.5)
- d. What type (and subtype) of multiplication or division problem is this an example of?



# Mr. Gomez's Cupcakes

Mr. Gomez had 20 cupcakes. He put the cupcakes into 4 boxes so that there were the same number of cupcakes in each box. How many cupcakes did Mr. Gomez put in each box?

- a. **Face Partner B:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.  
**Face Partner A:** How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b.  **Play: a student solves this problem** [smarturl.it/cm4.4](http://smarturl.it/cm4.4)
- c. What type (and subtype) of multiplication or division problem is this an example of?

# Coach Brown's Baseballs

Coach Brown has 72 baseballs. If 8 baseballs fit in a box, how many boxes can Coach Brown fill?

a. **Face Partner A:** Explain to your face partner why you don't want to use cubes to solve this problem with direct modeling.

**Face Partner B:** How else might a child go about solving this problem in a way that demonstrates making sense of the problem?

b.  **Play: a student solves this problem** [smarturl.it/cm4.2](https://smarturl.it/cm4.2)

c. What type (and subtype) of multiplication or division problem is this an example of?



# Which Unknown Names the Subtype?

| Problem                 | Given                | Unknown          | Subtype                           |
|-------------------------|----------------------|------------------|-----------------------------------|
| Grandma's cookies       | 3 groups, 6 in each  | product          | Equal Groups (Multiplication)     |
| Mr. Gomez's cupcakes    | 20 in all, 4 groups  | group size       | Equal Groups (Partitive Division) |
| Coach Brown's baseballs | 72 in all, 8 in each | number of groups | Equal Groups (Quotative Division) |

Quotative division is also called **measurement division**: we use the given size of each group to *measure* the total number of objects and see how many groups there are.

# Array Problems

**Definition 2.3.6**

An **Array problem** involves a set of objects arranged in a rectangular array.

Components: **number of rows**, **number of columns**, **product** (the total number of objects).

|                       |   |                          |   |                          |
|-----------------------|---|--------------------------|---|--------------------------|
| <b>number of rows</b> | × | <b>number of columns</b> | = | <b>product</b>           |
| <b>product</b>        | ÷ | <b>number of rows</b>    | = | <b>number of columns</b> |
| <b>product</b>        | ÷ | <b>number of columns</b> | = | <b>number of rows</b>    |

# The Three Array Subtypes

Table 2.3.8: Array Problem Subtypes

| Array (Multiplication)                                                                  | Array (Partitive Division)                                                            | Array (Quotative Division)                                                             |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| There are three rows of apples with five apples in each row. How many apples are there? | If 15 apples are arranged into three equal rows, how many apples will be in each row? | If 15 apples are arranged into equal rows of five apples, how many rows will there be? |
| $3 \times 5 = ?$                                                                        | $3 \times ? = 15$                                                                     | $? \times 5 = 15$                                                                      |
|                                                                                         | $15 \div 3 = ?$                                                                       | $15 \div 5 = ?$                                                                        |

# Why Does $5 \times 4 = 4 \times 5$ ?

A child reads Grandma's cookies as **three groups of six**, never six groups of three. The equal groups model is not **symmetric**. Arrays show children why the two products agree.



**5 rows of 4 chairs**



**4 rows of 5 chairs**

Same chairs, rearranged: the total does not change, so  $5 \times 4 = 4 \times 5$ .

# Area Problems

Definition 2.3.12

An **Area problem** involves representing the area of a rectangle using multiplication or division.

Rectangles only. Components: **length**, **width**, **product** (the area of the rectangle).

|        |   |        |   |        |
|--------|---|--------|---|--------|
| length | × | width  | = | area   |
| area   | ÷ | length | = | width  |
| area   | ÷ | width  | = | length |



# The Three Area Subtypes

Table 2.3.13: Area Problem Subtypes

| Area (Multiplication)                                    | Area (Partitive Division)                                                                               | Area (Quotative Division)                                                                              |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| What is the area of a three inch by five inch rectangle? | A rectangle has an area of 15 square inches. If the length is three inches long, how long is the width? | A rectangle has an area of 15 square inches. If the width is five inches long, how long is the length? |
| $3 \times 5 = ?$                                         | $3 \times ? = 15$                                                                                       | $? \times 5 = 15$                                                                                      |
|                                                          | $15 \div 3 = ?$                                                                                         | $15 \div 5 = ?$                                                                                        |



## Which Side Is the Length?

---

**A 3 inch by 5 inch rectangle:  
which side is the length?**

Either one. Length and width are interchangeable, so for Area problems the two division subtypes are too.

Table 2.3.13 tells them apart by the given side. **Length given:** Partitive Division.  
**Width given:** Quotative Division.

# Multiplicative Compare Problems

Definition 2.3.16

A **Multiplicative Compare problem** involves a comparison between two quantities in which one quantity is described as a multiple of the other quantity.

Components: **smaller** quantity, **bigger** quantity, **scale factor** (how many times larger).

|                 |   |                  |   |                  |
|-----------------|---|------------------|---|------------------|
| scale factor    | × | smaller quantity | = | bigger quantity  |
| bigger quantity | ÷ | smaller quantity | = | scale factor     |
| bigger quantity | ÷ | scale factor     | = | smaller quantity |

# Multiplicative Compare Subtypes

Table 2.3.17: Multiplicative Compare Problem Subtypes

| Multiplicative Compare<br>(Multiplication)                                                             | Multiplicative Compare<br>(Partitive Division)                                                       | Multiplicative Compare<br>(Quotative Division)                                                               |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| A blue hat costs \$5. A red hat costs 3 times as much as the blue hat. How much does the red hat cost? | A red hat costs \$15 and that is 3 times as much as a blue hat costs. How much does a blue hat cost? | A red hat costs \$15 and a blue hat costs \$5. How many times as much does the red hat cost as the blue hat? |
| $3 \times 5 = ?$                                                                                       | $3 \times ? = 15$                                                                                    | $? \times 5 = 15$                                                                                            |
|                                                                                                        | $15 \div 3 = ?$                                                                                      | $15 \div 5 = ?$                                                                                              |

# Write Three Problems From One Fact

---

Work with your group to use the math fact  $4 \times 3 = 12$  (or a related math fact) to create

- a. an “Array (Partitive Division)” problem.
- b. an “Area (Quotative Division)” problem.
- c. a “Multiplicative Compare (Multiplication)” problem.

Check each problem against its number facts: which quantity did you leave unknown?